IPCON® PET DET & FC

Detectable & Food Contact

Polyethylene Terephthalate Filament

This is the filament's Technical Data Sheet (TDS). If you want the Material Safety Data Sheet (MSDS), please contact the customer service to download.

• Product Information

Item	Data
Filament Diameter	1.75 ± 0.03 mm
Net Filament Weight	1 kg ± 0.5%
Filament Length	300 ± 2%
Spool Material	Cardboard (max heat resistance: 150 °C)
Spool Size	Outer Cardboard's diameter: 200 mm; Inner roller's diameter: 90 mm

• Filament's Properties

Physical Properties	Testing Methods	Data
Density	ISO 1183-3	1.39 g/cm³
Shrinkage Rate*	Specially-defined	0.28% - 0.51%
Surface Hardness	Shore hardness	78D
Saturated Water Absorption Rate	25 °C，55% RH，room air	0.37%
Melt Index	250 °C，2.16 kg	25.9 g/10 min
Melting Temperature	DSC，10 °C/min	235 °C
Vicar Softening Temperature	ISO 306，GB/T 1633	76 °C
Heat Deflection Temperature	ISO 75，0.45 MPa	71 °C

Mechanical Properties*	Testing Methods	Data*
Tensile Strength XY	ISO 527，GB/T 1040	33 MPa
Tensile Strength Z	ISO 527, GB/T 1040	17 MPa
Young's Modulus XY	ISO 527，GB/T 1040	1675 MPa
Young's Modulus Z	ISO 527，GB/T 1040	1454 MPa
Breaking Elongation Rate XY	ISO 527，GB/T 1040	14.7%
Breaking Elongation Rate Z	ISO 527，GB/T 1040	7.1%
Flexural Strength XY	ISO 178，GB/T 9341	63 MPa
Flexural Strength Z	ISO 178，GB/T 9341	41 MPa
Flexural Modulus XY	ISO 178，GB/T 9341	1752 MPa
Flexural Modulus Z	ISO 178，GB/T 9341	1667 MPa
Impact Strength XY	ISO 179，GB/T 1043	47.8 kJ/m²
Impact Strength Z	ISO 179，GB/T 1043	7.3 kJ/m²

*The shrinkage rate and mechanical properties were tested on specimens printed by IPCON using a Bambu P1S printer (nozzle temperature: 280 °C, bed temperature: 70 °C, printing speed: XY—220 mm/s and Z—70 mm/s) and IPCON PET DET & FC

(Blue) filament. The parameters of specimens of mechanical properties are as following: infill density—100%, infill pattern—concentric; and those of the 2 models of shrinkage (2 crosses that are vertical in the X and Y directions with designed lengths of 100.00 and 150.00 mm respectively) are as following: wall loop—2 and 6, infill density—15% and 25%, infill pattern—grid. The data is for user reference and comparison only. When using it, users are responsible for the safety, legal compliance and product performance.

• Main Features and Application Scenarios

1. Most models can be glued-free, detectable, food-contactable, specially designed for food, beverage and pharmaceutical manufacturing companies, water and oil resistant, hygienic and safe, high toughness, easy to print, supports high speed (max speed 300 mm/s), suitable for all mainstream printers, AMS and CFS.
2. Kitchen tools such as food jars, flour spoons, and pastry molds; food workshop stationery and tools such as pen casings, sealing clips, grabbing pliers, and ingredient boxes; food, beverage, and pharmaceutical processing equipment parts and other items that may come into contact with such products.

• Recommended Printing Settings

Drying Settings Before Printing	65 - 70 °C, 6 - 8 h (Blast drying oven)
Printing Temperature and Humidity	≤ 45 °C, ≤ 20% RH (Sealed with desiccant)
Storage Temperature and Humidity	≤ 35 °C，≤ 20% RH (Sealed with desiccant)
Compatible Support Material	Itself or support filament for PET
Compatible Printer Type	Open-frame, enclosed-frame (Suggest to open the front door and remove the upper lad)
AMS or CFS	Compatible
Compatible Nozzle Material	Stainless steel, hardened steel, tungsten carbide, diamond and other materials with very low risk of containing heavy metals (consult the nozzle manufacturer for details)
Compatible Nozzle Size	0.2 / 0.4 / 0.6 / 0.8 mm
Compatible Plate Type	Smooth / Textured PEI Plate or other plate
Plate Surface Preparation	Food contact use: do not use glue or apply food grade glue Non-food contact use: apply liquid glue or PVP solid glue stick
Nozzle Temperature	270 - 300 °C
Bed Temperature	65 - 80 °C
Fan Speed	10% - 40%
Max Print Speed	300 mm/s

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